Joshua Huang

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Education

Texas Tech University **August 2025 –**
PhD Student in Computer Science
Advised: Prof. Stas Tiomkin

The University of Texas at Austin **August 2021 – May 2025**
BS Electrical and Computer Engineering – GPA: 3.80

Experience

Research Assistant – Lubbock, TX **August 2025 –**
- Working on torque-aware and high-precision robotic control

DARPA ANSR Undergrad Researcher – Austin, TX **May 2024 – May 2025**
- Created partial Voronoi pathing generator from topological environment images
- Developed an image-to-MiniGrid program to allow for testing of proven RL models

General Motors Vision Research Intern – Austin, TX **May 2023 – August 2023**
- Patented approach for various methods to analyze sub-millimeter wear and structural stability of I-beams and double C-beams
- Focused on remedying 3 failure modes that have been extremely costly (Over $2,400,000 in revenue PER I-beam/double C-beam failure)
- Designed, tested, and refined a fully functional trolley system to collect data using a custom vision setup
- Demonstrated project with research/feasibility team to VP of North America Manufacturing, Manufacturing Executive Director - Trucks, VP of Global Manufacturing Engineering, and over a dozen more executives

General Motors Software Intern – Austin, TX **May 2022 – August 2022**
- Researched, created, and tested an analytical algorithm that takes live camera input and detects unwanted debris
- Extensive utilization of C, C++, and Excel — Performed data-driven research — Communicated and organized with team members in Ireland
- Debris detection algorithm will potentially save General Motors an estimated $10 million in production

Activities

Freelance Photographer – Self-employed **November 2022 –**
- Graduation photoshoots for UT Austin and A&M students
- Event photographer at UT Austin org events, A&M org events, retreats, Ascension Seton Austin Marathon, and sporting events

UT RecSports Staff – Photographer **August 2024 – May 2025**
- RecSports general activities, weightlifting, intramural sports, and sponsored events

FIRST FTC Robotics – Mentor **May 2021 – May 2024**
- Mentor of team 15 high school students during the competitive season
- Assists with design review, code debugging, and competitive strategy

Awards and Patents

Conveyor health inspection method **2026**
D. J. Carroll, C. A. Wiley, D. R. Kracko, N. S. Ellis, P. L. Griggs, E. R. Newcomb, R. Hardin, and **J. Huang**
U.S. Patent 12,754,143, Oct. 6, 2026. (Assignee: GM Global Technology Operations LLC)

FTC Texas Cup Winning Alliance Ranked 1st out of 358 Texas FTC teams **2021**

Skills

Programming Languages: Python, C/C++, Java,
Software: MuJoCo, IsaacLab/IsaacSim, PyTorch, Azure DevOps, Autodesk Fusion 360, PTC Creo 4.0/5.0, NX Siemens
Hardware: Fanuc
Interests: Photography, Mountain biking, Guitar, good engineering :)